

JEPAMatch: Geometric Representation Shaping for Semi-Supervised Learning

Ali AGHABABAEI, Aude SPORTISSE, Massih-Reza AMINI

Université Grenoble Alpes (UGA), Laboratoire d'Informatique de Grenoble (LIG), France

ECML-PKDD 2026

- **Deep learning** $\Rightarrow$ From academic research to the foundational technology of modern AI.
- Moved from **manual feature** engineering to learn hierarchical representation from **raw data**.
- Start with breakthroughs in:
 - **Computer Vision:** ResNet, ConvNet.
 - **Natural Language Processing:** Transformers, BERT.

- While deep learning consumes massive amounts of raw data, models are often starving for high-quality **annotated data** to achieve reliable performance.
- Acquiring this **supervised data** remains a critical bottleneck.
 - **The Human Cost:** Raw digital data is abundant, acquiring high-quality human annotations is expensive and slow.
 - **Domain Limitations:** In fields like medical imaging, labels require highly trained experts.
- This "**data wall**" severely restricts the scalability and accessibility of AI.

Semi-Supervised Learning (SSL): Leverages a small set of labeled data $\mathcal{X}_l$ to anchor the classes, and a vast pool of unlabeled data $\mathcal{X}_u$ to discover the data distribution.

- **Self-Training:** The model generates its own "pseudo-labels" for $\mathcal{X}_u$ and trains on them iteratively.
- **Consistency Regularization:** The model is forced to have the same output for different views, FixMatch¹.

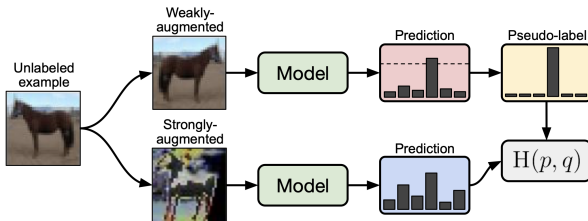


Figure 1: The FixMatch pipeline

¹K. Sohn et al., "FixMatch: Simplifying Semi-Supervised Learning with Consistency and Confidence", NeurIPS, 2020.

- **Confirmation Bias:** When the network reinforces its own early, incorrect predictions, it leads to dimensional collapse and poor generalization. Large batch sizes amplify confirmation bias.
- **Slow Convergence:** Millions of iterations are required to converge because the vast majority of unlabeled data is simply discarded in early training stages when confidence is low.
- **The Discriminative Bottleneck:** Existing frameworks rely exclusively on semantic decision boundaries, neglecting the underlying continuous geometry of the representation space.

Architectural Principle

JEPAMatch explicitly decouples discrete semantic classification from continuous geometric representation by Geometrical Self Supervised Learning.

- 1 Curriculum Level (Discrete):** Optimizes pseudo-labels using dynamic, class-aware thresholds.
- 2 Representation Level (Continuous):** Structures the latent space using an adaptive predictive architecture (adaptive LeJEPa).

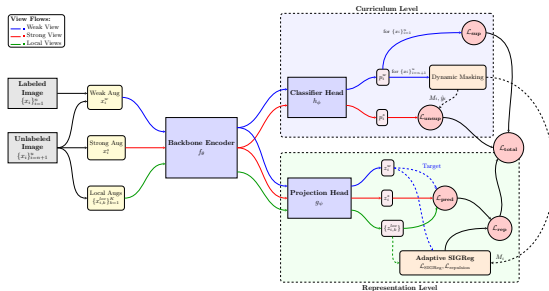


Figure 2: JEPAMatch architecture overview

JEPAMatch captures both semantic meaning and spatial topology. For each unlabeled image x_i , generates three types of views:

- **Weak global view** (x_i^w): Standard mild augmentations used for pseudo-labeling.
- **Strong global view** (x_i^s): Heavy perturbations to enforce invariance.
- **Local views** ($\{x_{i,k}^{loc}\}_{k=1}^K$): K small, structurally isolated crops capturing local dependencies within the image.



Figure 3: Global and local multi-view generation strategy

1. Curriculum Level: Pseudo-Labeling

- **FlexMatch:** Adaptive thresholds $\tau_c(t)$ dynamically filter weak views probability p_i to generate mask M_i :

$$\hat{y}_i^w = \arg \max(p_i^w)$$

$$M_i = \mathbb{1}[\max(p_i^w) \geq \tau_{\hat{y}_i^w}(t)]$$

- **Loss:** Cross-Entropy on the labeled set (B_l) with masked predictions on the unlabeled set (B_u):

$$\mathcal{L}_{\text{sup}} = \frac{1}{|B_l|} \sum_{i \in B_l} H(y_i, p_i^w)$$

$$\mathcal{L}_{\text{unsup}} = \frac{1}{|B_u|} \sum_{i \in B_u} M_i H(\hat{y}_i^w, p_i^s)$$

$$\mathcal{L}_{\text{CL}} = \mathcal{L}_{\text{sup}} + \lambda \mathcal{L}_{\text{unsup}}$$

2. Representation Level: LeJEP A²

- **Latent Shaping:** Shapes the latent representation.
- **Cross-View Prediction:** A projector predicts *global* views features using ONLY *local* representations. This enforces local-to-global semantic understanding.
- **SIG Regularization:** Embeddings are forced into an isotropic Gaussian $\mathcal{N}(0, I)$.

²R. Balestriero and Y. LeCun, "LeJEP A: Provable and Scalable Self-Supervised Learning Without the Heuristics", arXiv, 2025.

- **Prediction Loss:** Aligns local and strong views to the global context (where D is a distance metric and z_i is the representation output by the projection head.):

$$\mathcal{L}_{\text{pred}} = \frac{1}{|B_u|} \sum_{i \in B_u} \left(D(z_i^s, z_i^w) + \sum_{k=1}^K D(z_{i,k}^{\text{loc}}, z_i^w) \right)$$

- **SIGReg Loss:** Prevents dimensional collapse by matching the empirical characteristic function to an isotropic Gaussian:

$$\mathcal{L}_{\text{SIGReg}} = \frac{1}{K} \sum_{k=1}^K \text{SIGReg} \left(\{z_{i,k}^{\text{loc}}\}_{i \in B_u}, \mathcal{N}(0, I) \right)$$

fully expanded, uniform space.

- **Total Representation Loss:** Balances geometric invariance with structural regularity via β :

$$\mathcal{L}_{\text{rep}} = (1 - \beta) \mathcal{L}_{\text{pred}} + \beta \mathcal{L}_{\text{SIGReg}}$$

- **Adaptive Centering & Annealing:** Once pseudo-labels stabilize, we compute batch-wise class means μ_c . The target distribution shifts to these means, and the variance σ_t is annealed downwards to tighten intra-class clusters over time:

$$\mathcal{L}_{\text{SIGReg-Adp}} = \frac{1}{K} \sum_{k=1}^K \text{SIGReg}\left(\{z_{i,k}^{\text{loc}}\}_{i \in B_u}, \mathcal{N}(\mu_{\hat{y}_i}, \sigma_t^2 I)\right)$$

- **Active Repulsion Loss:** Prevents distinct class means from merging by enforcing a minimum distance margin m between them:

$$\mathcal{L}_{\text{repulsion}} = \frac{1}{C(C-1)} \sum_{c \neq c'} \max(0, m - \|\mu_c - \mu_{c'}\|_2^2)$$

- **Total Representation Loss:** Balances local-to-global prediction with adaptive geometric clustering:

$$\mathcal{L}_{\text{rep}} = (1 - \beta) \mathcal{L}_{\text{pred}} + \beta (\mathcal{L}_{\text{SIGReg-Adp}} + \mathcal{L}_{\text{repulsion}})$$

- JEPAMatch consistently achieves SOTA results, proving highly resilient in extreme low-label scenarios.
- **CIFAR-100 (4 labels/class):** Achieved a 34.25% **error rate**, severely outperforming FlexMatch (39.94%) and SoftMatch (37.10%).
- **Imbalanced Data:** Under severe class imbalance scenarios ($\gamma = 100$), JEPAMatch bypassed the Matthew Effect, beating baseline methods by 1.0% – 2.0%.

Table 1: Error rates on CIFAR-100 and Accuracy on Tiny-ImageNet against 2 baselines ³.

Method	CIFAR-100	Tiny-ImageNet (Acc %)		CIFAR-100 Imbalance (Err %)		
	400 labels (Err %)	1k labels	10k labels	$\gamma = 20$	$\gamma = 50$	$\gamma = 100$
FlexMatch ³	39.94	58.27	72.11	49.11	57.20	62.70
SoftMatch ⁴	37.10	59.91	74.08	48.09	56.24	61.08
JEPAMatch	34.25	61.18	75.50	46.27	55.16	59.93

³Zhang, et al. "Flexmatch: Boosting semi-supervised learning with curriculum pseudo labeling", NeurIPS, 2021.

⁴Chen, et al. "Softmatch: Addressing the quantity-quality trade-off in semi-supervised learning", ICRL, 2023.

- **Speedups:** Reached same accuracy 50k iterations earlier than FlexMatch. (JEPAMatch process increase training time by 1.8)
- **Unlabeled Data:** Utilization unlabeled data from FlexMatch to JEPAMatch. More passed τ .

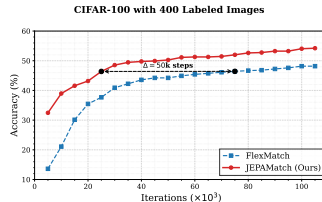


Figure 4: Convergence speed compared to baselines

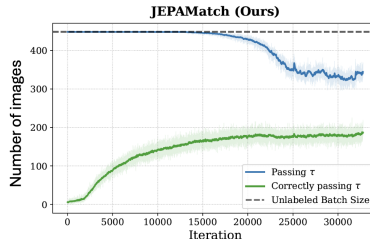
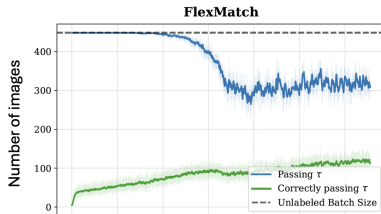


Figure 5: data-utilization

- **Pseudo-Labeling Accuracy:** JEPAMatch maintains a stable, upward trajectory from early stages, reaching $\sim 50\%$ true label accuracy on CIFAR-100 compared to FlexMatch's $\sim 35\%$.
- **Filtering Noise:** The integration of global and local JEPA representations provides robust filtering of noisy pseudo-labels.

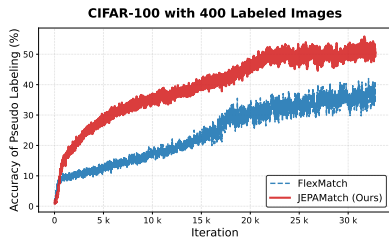


Figure 6 (a): Pseudo-labeling accuracy (CIFAR-100)

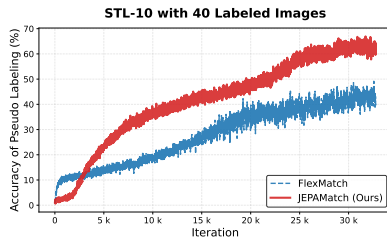


Figure 6 (b): Pseudo-labeling accuracy (STL-10)

- **Baseline Failure:** The major class can receive up to 20% of all pseudo-labels, demonstrating massive class dominance.
- **Balanced Discovery:** Adaptive Class-Wise SIGReg enforces balanced spatial allocation, preventing head classes from dominating the geometric space near 7%.

CIFAR-100 with 400 Labeled Images

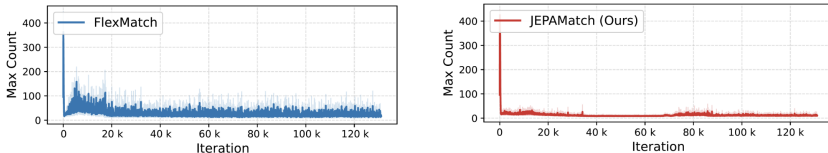


Figure 7: Evolution of the maximum class count during training

The Bottleneck: Data Annotation

- Modern deep learning achieves remarkable performance, but at a growing cost in **supervision**; confidence-based pseudo-labeling (FixMatch) suffers from majority-class dominance and confirmation bias.

JEPAMatch: Decoupled Learning

- Separates a discrete **Curriculum Level** (dynamic pseudo-labeling) from a continuous **Representation Level** (LeJEPA-inspired adaptive class-wise SIGReg), preventing dimensional collapse.

Results & Impact

- SOTA accuracy on CIFAR-100, STL-10, and Tiny-ImageNet with $8\times$ fewer training iterations; up to 2% gains under severe class imbalance.

Future Directions

- Scale to multimodal and NLP settings, and further investigate representation shaping under class imbalance.

Thank You for Your Attention

Université Grenoble Alpes — Laboratoire d'Informatique de Grenoble

Questions & Discussion